

Water and Power use for AI Data Centers?

Posted by u/Ok_Tangelo_6070 • 0 points • 9 comments

One thing that I'm curious about is this...

1. Where are they going to get all of the water and power to run the hundreds of data centers that they plan to build?

They can't even get enough RAM chips to build these things...

Comments

u/Mabuse046 • 1 points • 2026-02-02 03:09 UTC

Sorry to be that guy but our planet is like 90% water. The materials needed to make RAM exist in much smaller quantities. So... apples to oranges.

u/DimphoPopi • 1 points • 2026-02-02 04:14 UTC

You need de-ionised water for cooling as salt content in seawater would damage parts, kinda like the gunk that builds up on your kettle over time. It's easier to deionise from fresh water than desalination of salt water is more expensive, and you would build next to the ocean not in the middle of a fresh water source if you use salt water.

Cooling using seawater is common in the nuclear industry, de ionised water is used to cool systems that cannot be used again, it is radioactive and in tech, water evaporated and the remaining contaminated water cannot be used again, even for irrigation purposes.

You could argue that the vapour will make rain, however due to climate change and global warming, water systems are imbalanced, meaning it usually leads to drought in drought-prone areas and to floods in floods prone area. So hurricanes in hurricane areas become worse and drought in drought areas becomes worse.

u/Mabuse046 • 1 points • 2026-02-02 04:29 UTC

I actually do argue that it will produce rain. The Earth is a closed system. There's nowhere else to go. And while I don't dismiss global warming, I think you might be under a false impression that the current water distribution is natural and that the water redistributing itself is a consequence of our meddling. The truth is the other way around. Humans took marshes and deserts and turned them all into homogeneous cities. For instance, you get that Las Vegas was just a desert, right? So when you tell me that drought prone areas will become drier or flood prone areas will become wetter,

that sounds to me like the water is going back to where it was supposed to be in the first place. i.e. Nature is healing.

u/khisanthmagus • 1 points • 2026-02-03 15:08 UTC

If you drain the aquifer in Area A to cool datacenters and it then rains in Area B, a couple hundred miles downwind, I don't think that Area A is going to be comforted that it is fine that the datacenter used all that water because it just rained back down.

u/Mabuse046 • 1 points • 2026-02-03 16:18 UTC

An area with an aquifer you don't really have to worry about that. The presence of an aquifer proves that the area naturally receives rain. The important thing to remember about evaporative cooling only works in dry environments where evaporation can happen rapidly. I live in the Midwest and I've never even seen a evaporative / swamp cooler. So the people in area A do not have an aquifer, they are having water pumped in from area B. If the water evaporates in area A, it will naturally rain back down on area B, where it came from. So the almost ironic question behind it is what your stance is on human beings manipulating the natural order of things for the sake of human comfort.

u/DimphoPopi • 1 points • 2026-02-02 04:42 UTC

That is an imbalance as the environment becomes uninhabitable not only for humans but for animals, birds, and insects. A climate balance allows Earth to regenerate resources but if these conditions become extreme, the environment where there are floods cannot generate trees, animals leave or die and the ecosystem does not function, in the same way that a drought causes the same results. The environment and ecosystem depend on this delicate dance, extremities are detrimental which is what humans have been doing since the industrial revolution, and not AI is worsening conditions both in emissions, water depletion, habitat destruction, and pollution.

u/Mabuse046 • 1 points • 2026-02-02 05:07 UTC

The environment and ecosystem depended - past tense - on this. We're not living in a natural ecosystem. Humans have changed the natural landscape of the planet and driven many, MANY species to extinction. The very species that were adapted to live in those environments before humans decided we needed to reshape the world to make it more comfortable for us. To put things in perspective, this is like claiming that if your air conditioner breaks your house will become unnaturally warm. But that's wrong because warm is the natural state and it's the air conditioning that's unnatural.

u/DimphoPopi • 1 points • 2026-02-02 11:23 UTC

I think I get your point; this is a true diagnosis of the current human world. We have broken the biosphere, even though co-existing is way more beneficial for humanity. That is true. The effects, however is that now Earth will not support our human-made world.

Droughts, floods, heatwaves, resource scarcities, hurricanes, diseases (from pollution, pests, melting icecaps, bacteria), all will be unstoppable, a response for breaking the biosphere. Many people, especially in poorer countries, will die. But yeah, you can't use saltwater sources, fresh water sources are ideal.

u/Mabuse046

• 2 points
• 2026-02-02 11:31 UTC

I hate to be a doom sayer but we're not likely to survive long regardless. Current weather projections suggest that as early as 2050, centered around Arizona, we should begin seeing more

people die per year of heat related causes than traffic accidents. And to our knowledge this is not reversible. Our only hope is in some technological breakthrough. Maybe it's AI and maybe it's not, but this isn't the time to be holding back on any possible way forward.

u/bag_of_drums • 3 points • 2026-02-01 19:09 UTC

pretty sure they're just gonna keep building them in places with cheap power and water until those resources run out, then act surprised when the grid can't handle it

u/Ok_Tangelo_6070 • 2 points • 2026-02-01 19:15 UTC

Yeah.